

Home WiFi Network Lab Report

Cisco Packet Tracer Project Documentation & Configuration Guide

Project File Name: HomeWiFiNetwork.pkt | **Subnet:** 192.168.10.0/24 | **Default Gateway:** 192.168.10.1

Introduction & Objective

This technical report documents the complete step-by-step procedure for designing, configuring, and verifying a wireless home local area network (WLAN) using Cisco Packet Tracer. The objective is to interconnect three Personal Computers (PCs) with a wireless router, assign appropriate static IPv4 addressing within the same subnet, establish default gateway routing, and successfully test network connectivity using ICMP ping commands.

Step 1: Network Topology & Creation (HomeWiFiNetwork.pkt)

The initial phase involves launching Cisco Packet Tracer, setting up the virtual workspace, and deploying the required hardware components to form the topology.

- 1. Open Cisco Packet Tracer:** Launch the application and start a fresh blank project.
- 2. Select and Place Devices:**
 - Navigate to the device selection toolbar and select **Wireless Devices** -> choose a wireless router model (e.g., *WRT300N* or *Wireless Router*).
 - Navigate to **End Devices** -> select **PC** and place three distinct PC units onto the logical workspace (labeled PC0, PC1, and PC2).
- 3. Physical/Wireless Interconnection:**
 - By default, Cisco Packet Tracer PCs come with an Ethernet NIC (FastEthernet0). To connect them wirelessly to the wireless router, replace the FastEthernet module with a wireless NIC (e.g., **WMP300N**) or use direct Ethernet patch cables if simulating a hybrid wired/wireless setup. For this project, wireless modules were installed on all three PCs to connect via SSID.
 - Save the project file under the exact name: HomeWiFiNetwork.pkt.

Logical Network Topology Diagram

[PC0] (192.168.10.10) -- \
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[PC1] (192.168.10.11) ---- (Wireless Router) [192.168.10.1]
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[PC2] (192.168.10.12) -- /

Wireless Router (192.168.10.1)

PC0 (192.168.10.10)

PC1 (192.168.10.11)

PC2 (192.168.10.12)

Step 2: IP Addressing & Configuration

To ensure all devices can communicate seamlessly, they must reside on the same IP subnet (192.168.10.x). The wireless router acts as the default gateway and DHCP/Access Point hub.

Router Configuration

1. Click on the **Wireless Router**, navigate to the **GUI / Setup** tab.
2. Set Local IP Address to 192.168.10.1 with Subnet Mask 255.255.255.0.
3. Configure Wireless settings (SSID: HomeNetwork, Security: WPA2-PSK with a passphrase).

Static IP Allocation Table

Device Name	Interface	IP Address	Subnet Mask	Default Gateway
Wireless Router	Vlan1 / Gateway	192.168.10.1	255.255.255.0	N/A (Self)
PC0	Wireless0 / FastEthernet0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	Wireless0 / FastEthernet0	192.168.10.11	255.255.255.0	192.168.10.1
PC2	Wireless0 / FastEthernet0	192.168.10.12	255.255.255.0	192.168.10.1

Configuration Steps on each PC: Click PC -> **Desktop** tab -> **IP Configuration** -> Select *Static* -> Enter IPv4 Address, Subnet Mask, and Default Gateway as specified above.

Step 3: Connectivity Testing & Verification

Network connectivity was verified using the built-in Packet Tracer **Command Prompt** tool on each PC to execute ICMP ping tests against neighboring nodes and the router.

Test Execution: PC0 (192.168.10.10) Ping Log

```
C:\> ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Ping statistics for 192.168.10.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

C:\> ping 192.168.10.11
Pinging 192.168.10.11 with 32 bytes of data:
Reply from 192.168.10.11: bytes=32 time=1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.10.11: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

C:\> ping 192.168.10.12
Pinging 192.168.10.12 with 32 bytes of data:
Reply from 192.168.10.12: bytes=32 time=1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.10.12: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Step 4: Documentation & Configuration Summary

All configuration windows, IP settings, and ping test results confirm that the home wireless network was successfully deployed:

- **Project Save:** The project was saved successfully as `HomeWiFiNetwork.pkt`.
- **IP Verification:** Static IP addresses were verified using `ipconfig /all` on each PC terminal.
- **Full Mesh Connectivity:** 100% successful packet delivery was observed across all nodes (PC0, PC1, PC2, and Router).